

Human arm impedance estimation in dynamic human-robot interaction, a preliminary study

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1. Introduction

The estimation of the human impedance interacting with a physical environment provides modelling insights both for the field of human movement science (Palazzolo et al. 2007) and for the design of innovative controllers for collaborative robotics based on physical human-robot interactions (Tsuji and Tanaka 2008).

The mechanical impedance relates restitution forces to kinematics. Elementary models, such as spring-damper-mass (KBM), have been used as linearisation for small deviations because they could approach human behaviour (Tsuji and Tanaka 2008).

To achieve mechanical impedance estimations, one must perturb a movement while measuring both force and kinematics. The virtual trajectory, that is the unperturbed trajectory, must be either known or estimated. The estimation can either be done at the joint level or at the endpoint level to observe a global behaviour (Erden and Billard 2015).

In most of the literature, the estimations are conducted in well mastered conditions (Burdet et al. 2000; Erden and Billard 2015), that might be complicated to replicate outside laboratories.

Indeed, in contexts outside of labs, the virtual trajectory is often unknown or might not be approximated using previous trajectories because the repeatability is not sufficient to reach a decent precision. Indeed, the perturbations must remain small on the one hand not to modify the task, and on the other hand, because the linear impedance model stands for slight deviations.

In previous work, Avrin et al. (2018) studied a benchmark task, the ball-bouncing task. Fortineau et al. (2020, 2021) proposed an experimental testbed and identification methodology. This work tackled impedance estimation in a task where average trajectories could not be used and virtual trajectories were unknown.

Here, a preliminary experimental case study conducted on three different users tends to demonstrate the ability to observe stiffness variations in a dynamic task using the described methodology, with several users, extending the validity of the method.

2. Methods

2.1 Experimental protocol

Here we briefly present the testbed designed for human arm impedance estimation; for further details, please refer to Fortineau et al. (2020).

A user is manipulating an admittance-controlled poly-articulated robot, manoeuvring its endpoint as a physical interface, to control a paddle simulated in a digital environment. The motion is limited to the vertical plane since the ball-bouncing task is unidimensional. The user has to move the paddle to bounce the ball to a target height that settles the task's frequency.

The robot introduces short (<60ms) force perturbations (~5N) at the joint-control level to generate dynamic deviations. The perturbations have a random direction and have stochastic timing pre-programmed for three different phases of the cyclic pattern (c1, c2, c3), that can be seen on Fig.1.

Three healthy participants (u_1 , u_2 , u_3) volunteered to participate in the preliminary trial. They are all members of the authors' laboratory and are already sensitive to robot manipulation. However, only one of the subjects, u_1 , is considered an expert for spending more than one hour manipulating the robot, while the two others only had approximately 10 minutes of training with the robot before the recording.

They were aware of haptic feedback and were asked to do their best to bounce the ball to a given target height while trying to maintain a cyclic behaviour.

The ethical committee of Université Paris-Saclay, Polethis, approved the study protocol.

2.2 Methodology

For the sake of brevity, the identification method is not reminded here; please refer to Fortineau et al. (2021) for details.

The impedance estimation is sensitive to errors in the trajectory estimation of both virtual force and position; outliers must be discarded. We used the coefficient of determination R^2 of the reconstructed position from identified parameters, compared to measurements, to discard data with $R^2 < 0.5$. Stiffnesses outside five scaled absolute median deviation were also considered to be outliers.

3. Results and discussion

The three users, u_1 , u_2 , u_3 , were recorded, respectively, for 10min50s, 5min30s and 8min10s, and were capable of maintaining a cyclic behaviour during the experiment. After clearing the outliers, 113 (46, 36, 31), 53 (25, 15, 13) and 78 (27, 34, 17) identifications were spread among the three phases (indicated in parentheses), as shown on Fig. 1, with dots.

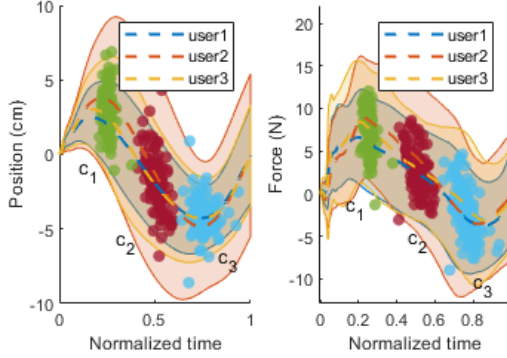


Fig. 1: Average and standard, position and force cycles

Fig. 2 shows the distribution of the estimated stiffness against the cycle frequency according to both phases and users. We can see that for the experimented user, each cycle's medians are closer than for the other users, which might indicate expertise. The highest median stiffnesses are recorded for the second user, and the first phase (after the ball impact) has the most important median among the three users.

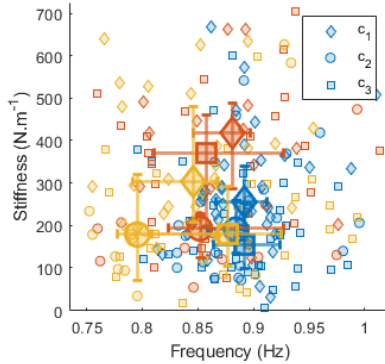


Fig. 2: Estimated stiffness against cycle frequency, colours are respectively blue, orange and yellow according to the user and the marker shape indicates the class. The bars and bigger markers indicate the quartiles.

Within user				Between user			
	C1/C2	C1/C3	C2/C3		u1/u2	u1/u3	u2/u3
u ₁	ns	s	ns	C ₁	ns	ns	ns
u ₂	s	ns	ns	C ₂	ns	ns	ns
u ₃	ns	s	ns	C ₃	s	ns	s

Table 1: Significance paired student tests

Results from Fig. 2 must be compared using Table 1, which unveils the significance differences, using 18 Student tests with a probability level of 0.003 (5% with Bonferroni correction). It can be observed that no significant differences were observed between the first and last user. One significant difference was found for each user within cyclic phases.

4. Conclusions

The results presented here comforts the findings of previous work. It is interesting to underline that during the experiment, the second user spontaneously declared that he felt “too stiff”, while the third user said that he “almost forgot about the robot”, which indicates a more relaxed behaviour. These remarks are bored out by the estimated median stiffness.

However, these encouraging results need to be tested on a more significant number of users to limit the influence of the variance in stiffness estimation; in order to be able to draw meaningful conclusions for the human-robot physical interactions.

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